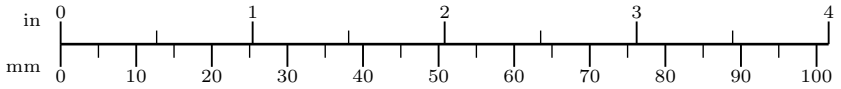
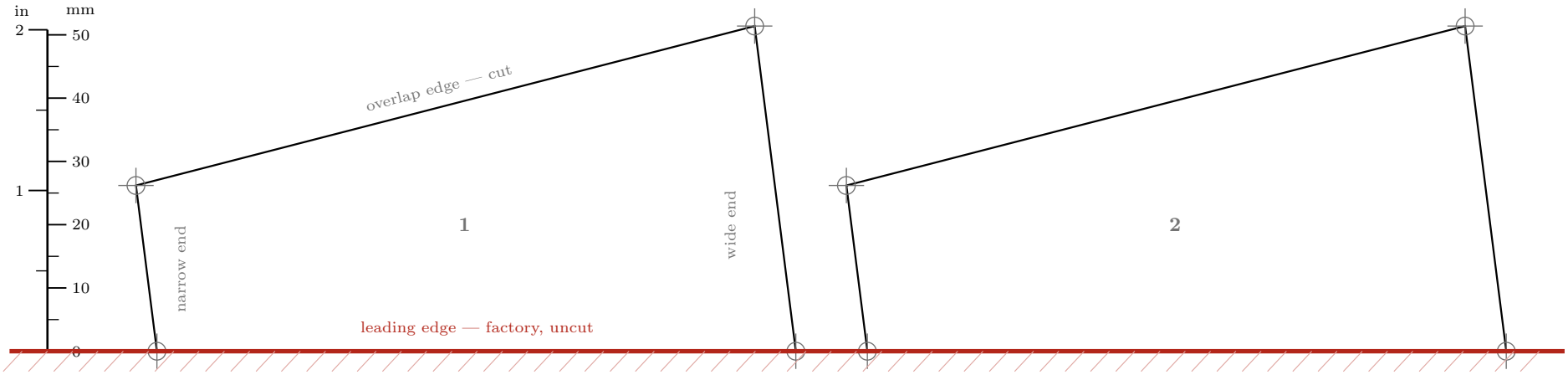
Kaleidoscope mirror cutting template

100 mm prototype · three identical mirrors · drawn 1:1

Print at 100%. In the print dialog choose *Actual size* or *100%* — not *Fit to page*, not *Shrink oversized pages*. Check *both* rulers below with a real ruler before cutting anything. If they are off, so is every dimension here.



0–100 below = 100.0 mm exactly
0–4 above = 4.000 in exactly



FACTORY EDGE OF THE TILE — lay this line on it. Never cut or sand along it.

Everything below the line hangs off the tile, which is how you sight the alignment.

Blank: parallel edges **26.40** and **51.80 mm**, both slanted edges **101.07 mm**, height between the parallel edges **100.27 mm**. Three needed; two are drawn. Cut the third the same way against another factory edge — a 298.45 mm edge takes two blanks and the tile has four edges.

Why the factory edge. Only one edge of each mirror is optically critical: the leading edge, which lands on the neighbouring mirror at the seam. Its coating has to run right to the edge, and no cut you make beats the edge the tile came with. The overlap edge is buried in dead space behind its neighbour and never sees light, so it only needs to be free of burrs. The two ends hide under the end caps.

Placing it. Leave the protective film on. Lay the sheet so the heavy line sits on a factory edge, with the strip below it overhanging so you can sight the alignment, and tape it down. Prick through the centre of all eight circled marks with a pin, lift the paper, join the pricks with your straightedge, and cut only the three non-factory edges.

Coated side up. These are rear-surface mirrors — clear plastic in front, reflective coating on the back. A jeweller's saw cuts on the downstroke, so the teeth enter the top face and break out of the bottom; coated side up puts that breakout on the clear face. Blade: a #2, teeth pointing toward the handle.

Mark the factory edge on the coated back as soon as each blank is free — a stripe of marker is enough. Once the four edges are cut apart it is easy to lose track of which one it was, and at assembly it has to end up on the corner-block side of its rail.

Width is forgiving. Anything from 25.5 to 27.4 mm across the narrow edge will seat, because the surplus buries itself behind the neighbouring mirror. Err generous and sand back. Sand only the cut edges, never the factory edge: 400-grit silicon carbide to take the burr off, 600 to finish.